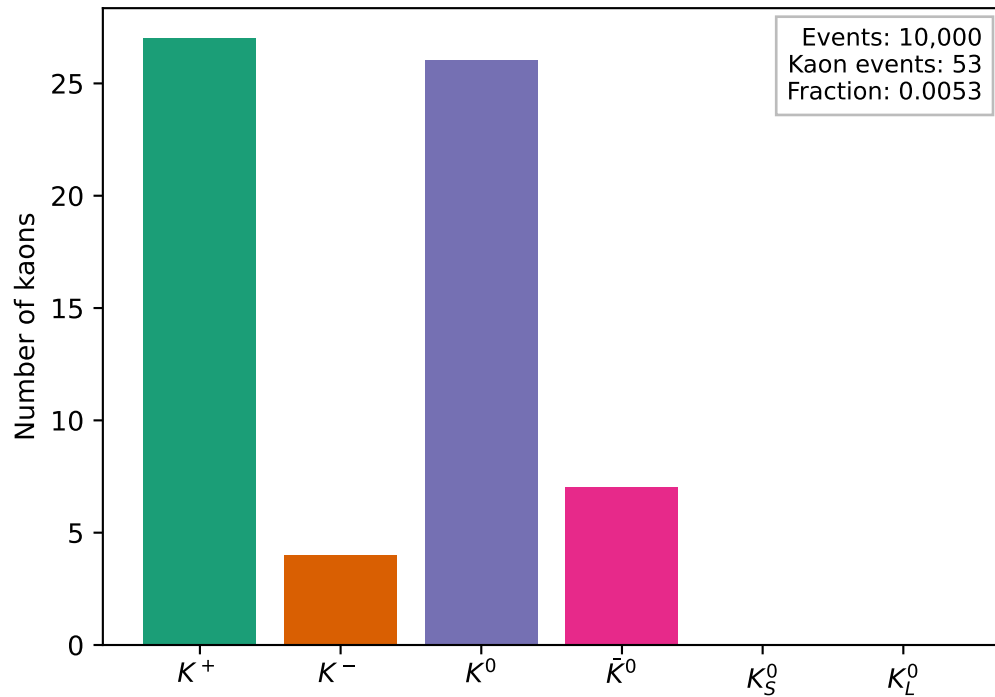
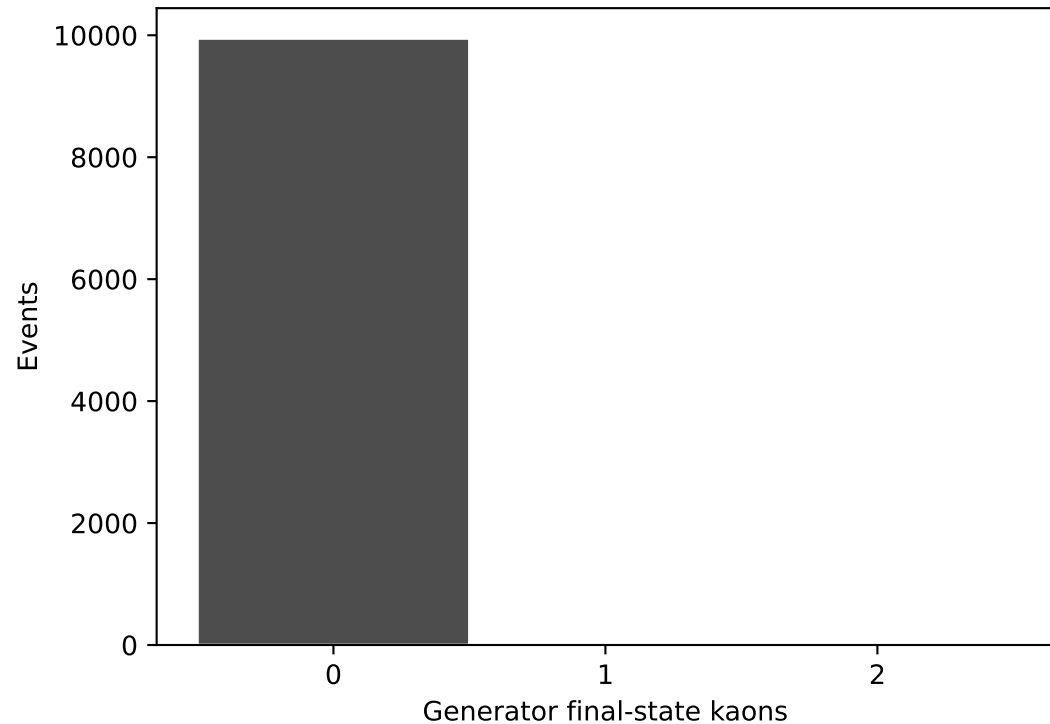


Atmospheric-neutrino kaon truth overview

Generator final-state kaon species

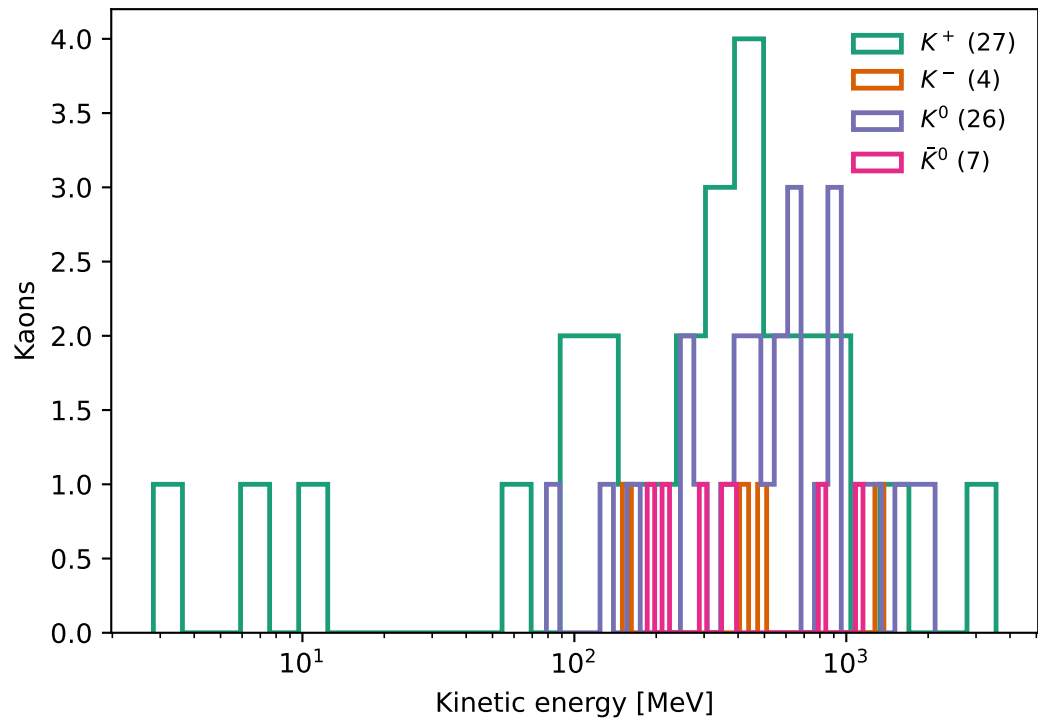


Kaon multiplicity per event

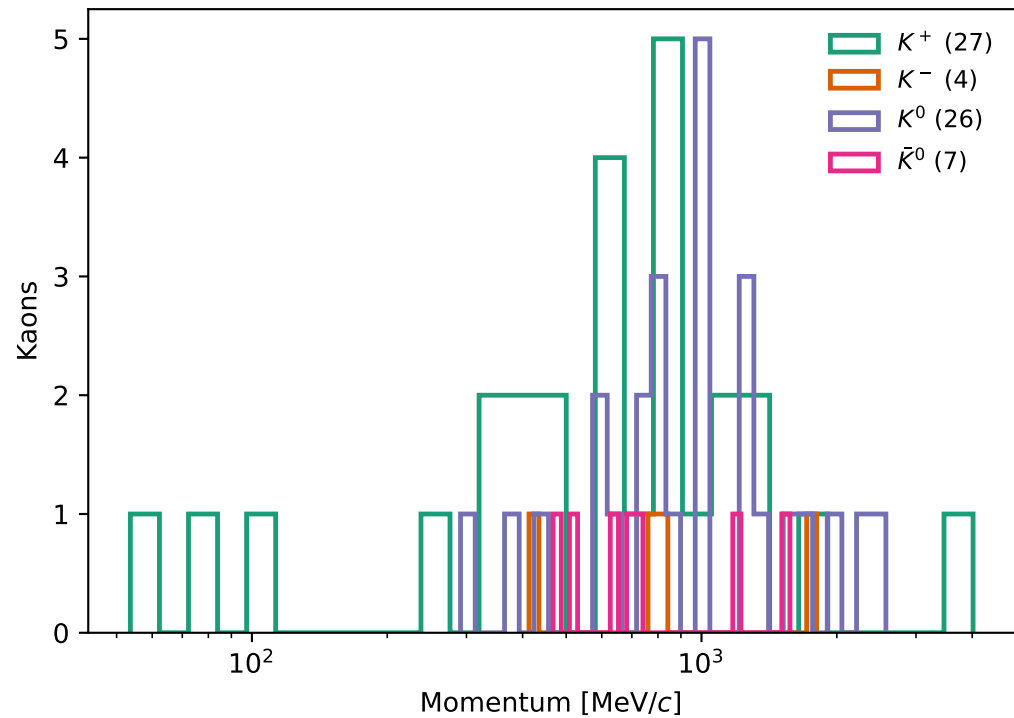


Generator final-state kaon kinematics

Kaon kinetic energy

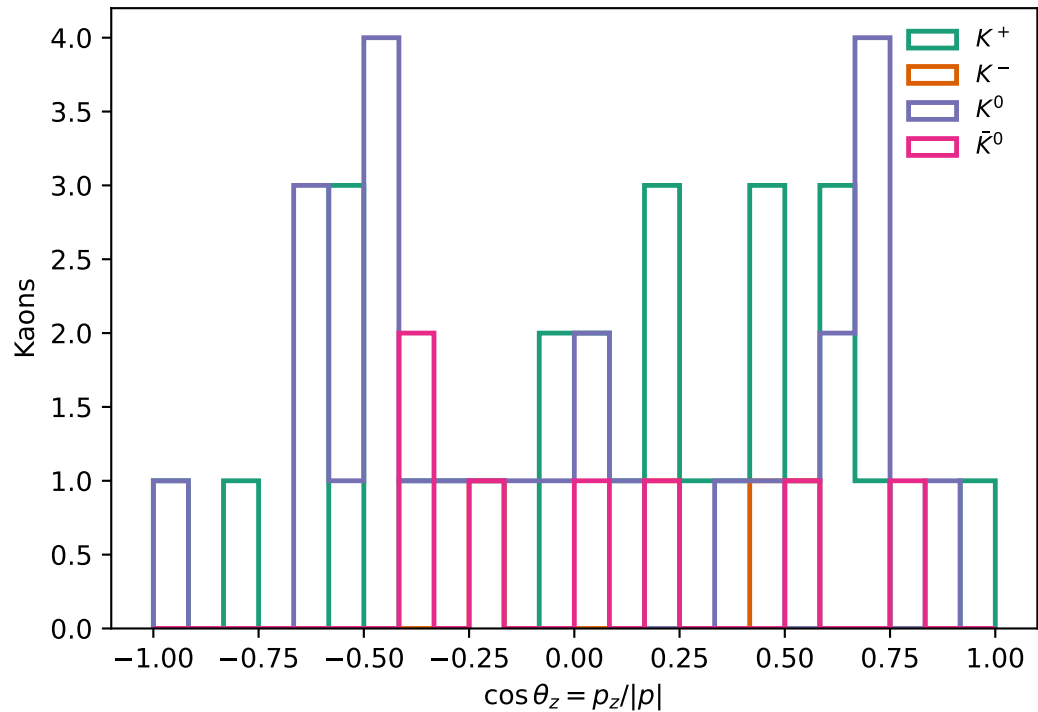


Kaon momentum

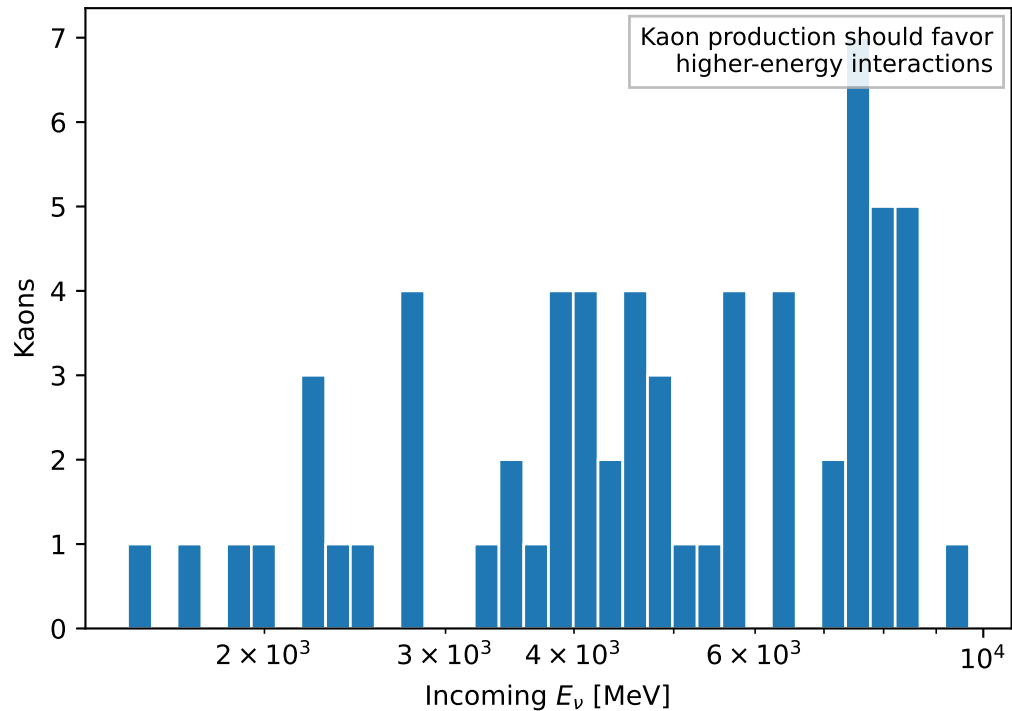


Kaon direction and production energy

Kaon direction

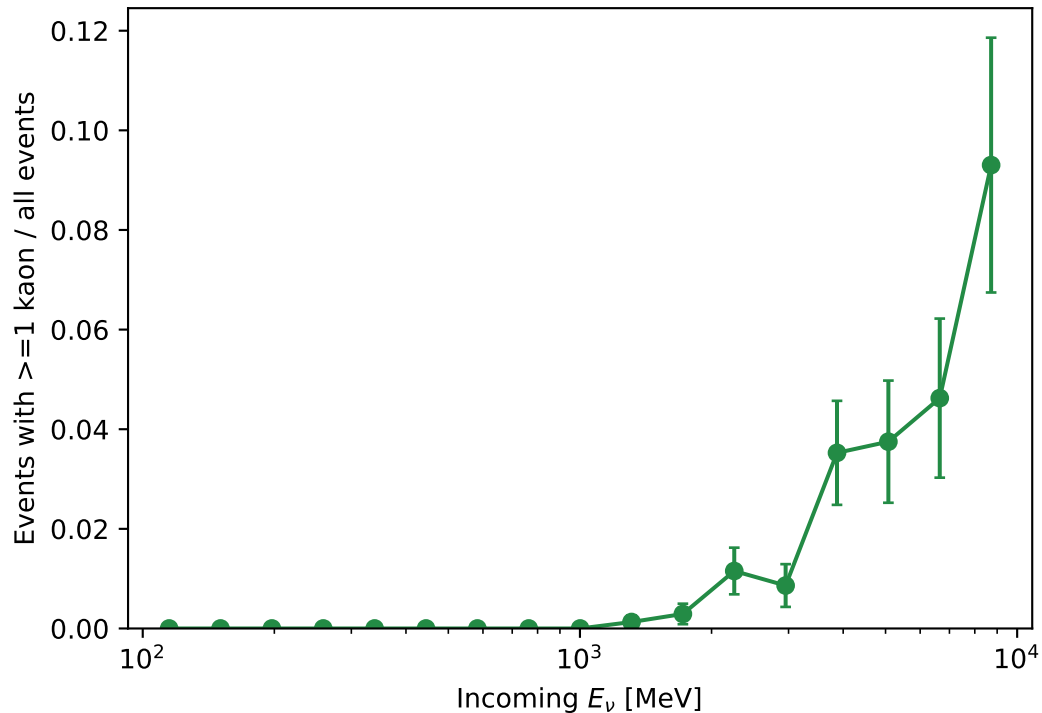


Incoming energy for produced kaons

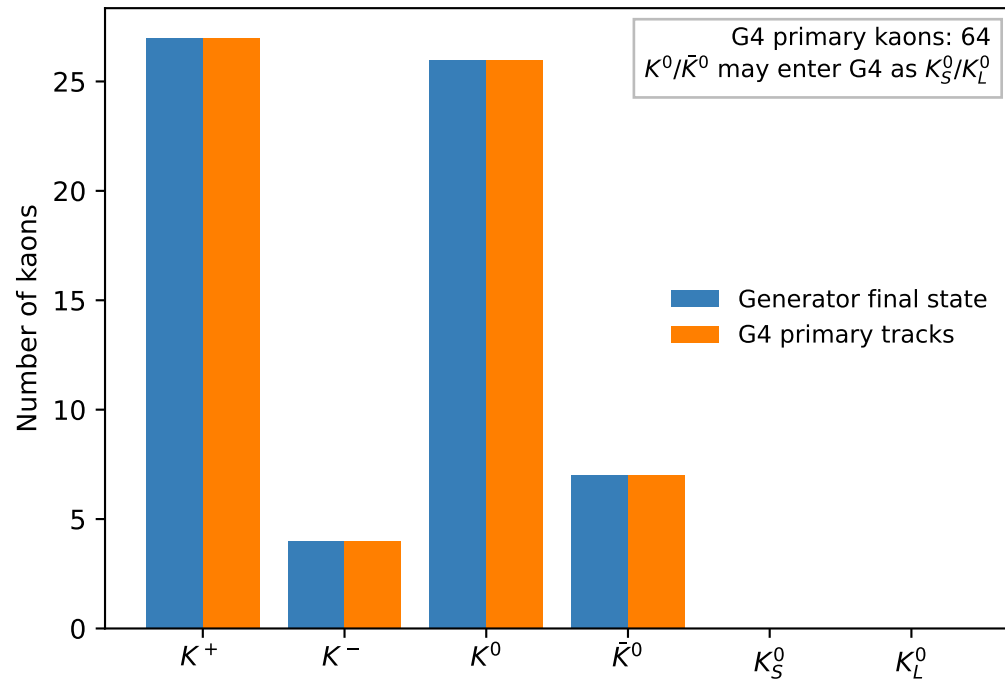


Kaon production trend and transport handoff

Kaon-event fraction vs energy

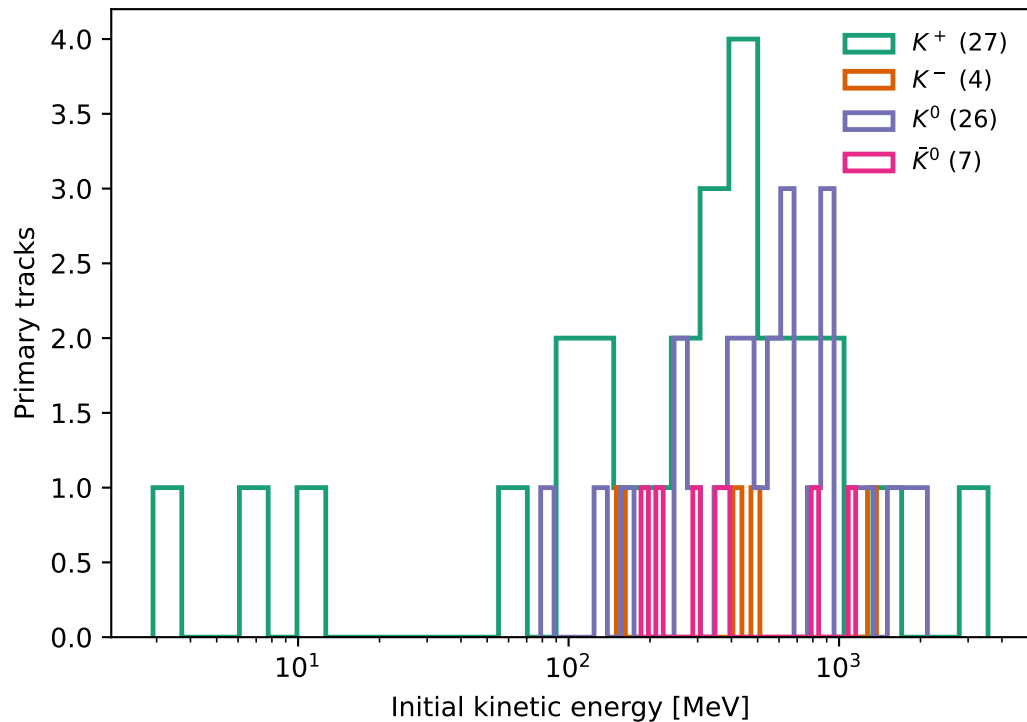


Generator-to-transport bookkeeping

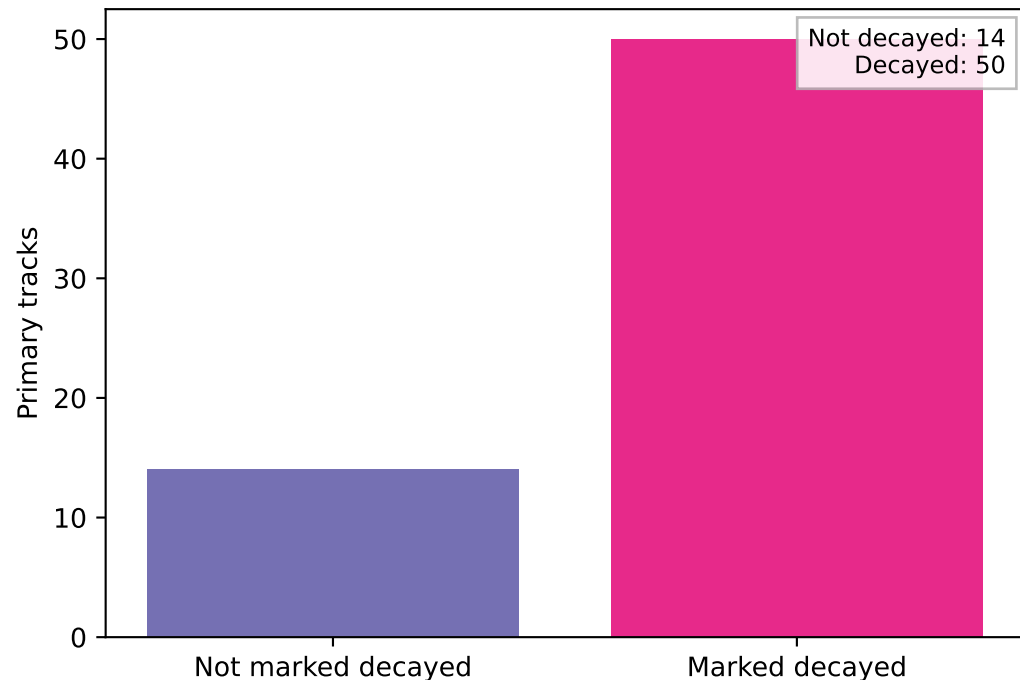


Geant4 primary-kaon transport checks

Transported primary-kaon energy



Transported primary-kaon decay flag



Kaon sanity-check report

What 'generator final state' means

The generator initial state contains the incoming atmospheric neutrino and argon target. GENIE models the neutrino-argon interaction, including hadron production and intranuclear final-state interactions. The generator final state is the particle list that exits that generator-level nuclear system and is handed to Geant4. The first-page bars therefore count kaons produced by the interaction model, before Geant4 propagates, scatters, or decays them; they are not hit counts.

Sample accounting

The 10,000-interaction sample contains 64 generator final-state kaons in 53 events, an event fraction of 0.530%. There are 31 charged and 33 neutral-flavor kaons. The species counts are $K^+=27$, $K^-=4$, $K^0=26$, and $\text{anti-}K^0=7$. The K^+/K^- ratio is 6.75.

Why the species pattern is plausible

Rare kaon production is expected at these energies because creating strange hadrons requires more available hadronic energy than quasielastic scattering or single-pion production. K^+ exceeding K^- is also qualitatively reasonable in a neutrino-rich sample: associated strangeness production can produce K^+ together with a hyperon, whereas K^- production is more constrained and K^- has stronger absorption channels in nuclear matter. Exact ratios remain strongly dependent on GENIE's hadronization and final-state-interaction model.

Multiplicity interpretation

There are 11 kaons beyond a one-kaon-per-kaon-event baseline, so a small number of events contain multiple kaons. Most interactions contain no kaon, as expected for an inclusive atmospheric sample. The very small selected population means individual-bin fluctuations are large and smooth theoretical shapes should not be expected from only 64 kaons.

Kaon physics interpretation

Production energy and kinematics

The median kaon kinetic energy is 438 MeV, with a central 90% interval of 63 to 1573 MeV. The incoming neutrino energy for produced kaons has a median of 4940 MeV and begins near 1472 MeV in this finite sample. The rising kaon-event fraction with neutrino energy is the expected threshold and phase-space behavior: higher-energy interactions can support strange-hadron production more readily.

Direction distribution

The kaon $\cos(\theta_z)$ values populate both signs and do not show an unphysical spike at one coordinate boundary. With only 64 particles, the jagged appearance is statistical. A quantitative atmospheric zenith comparison requires the flux-axis convention, oscillation weights, and substantially more kaon statistics.

Generator-to-Geant4 handoff

The generator contains 64 kaons and Geant4 records 64 primary kaon tracks, with matching displayed species totals in this file. This is an excellent bookkeeping check: no produced kaons appear to have been dropped or duplicated at transport initialization. In other configurations, neutral K^0 and anti- K^0 flavor states may be represented as physical K-short or K-long states, so neutral totals are safer than requiring every neutral PDG bar to match.

Kaon transport assessment

Transport outcome

Geant4 marks 50 of 64 primary kaons (78.1%) as decayed and 14 as not decayed. This is plausible for kaons traversing a finite liquid-argon volume: the result reflects competition among decay, hadronic interactions, stopping, and escape. The flag alone does not classify the non-decayed tracks by absorption versus leaving the active volume.

Conclusion and limitations

The rarity, increasing production fraction with energy, K^+ dominance, broad kinematics, and exact generator-to-primary-track accounting are all physically sensible. The main limitation is statistics. A quantitative kaon-production validation should use a larger weighted sample and compare GENIE tunes, hadronization/FSI variations, and available neutrino-induced strangeness data.

Recommended quantitative follow-up

Increase the generated exposure until each important kaon species has enough entries for stable binned comparisons. Then preserve the event weights and compare kaon multiplicity, momentum, neutrino-energy dependence, and strange-baryon associations across GENIE tune and FSI variations. Treat the present plots as a successful bookkeeping and qualitative-physics check, not a precision strangeness result.